

K2-29b

R-2253

Benchmark run · workflow W8-benchmark-deep · record deep-bench_K2-29_210212 · created 2026-08-16T00:16:39+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459257.6236 +/- 0.0044 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.055 +/- 0.044
Transit depth (Rp/Rs)^2	0.3 +/- 0.48 [%]
Orbital Inclination (inc)	86.87 +/- 2.39
Airmass coefficient 1 (a1)	1.302 +/- 0.013
Airmass coefficient 2 (a2)	-0.194 +/- 0.0073
Scatter in the residuals of the lightcurve fit is	0.96 %
Best Comparison Star	None
Optimal Aperture	3.95
Optimal Annulus	7.51
Transit Duration (day)	0.04 +/- 0.018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.055 ± 0.044 vs 0.1419 (deviation 1.93σ)
Duration fit vs published	0.04 d vs 0.0925 d (deviation 2.86σ)
Mid-time O–C	-4.7 min
Depth SNR	1.2
Residual scatter	0.96 %

Light curve

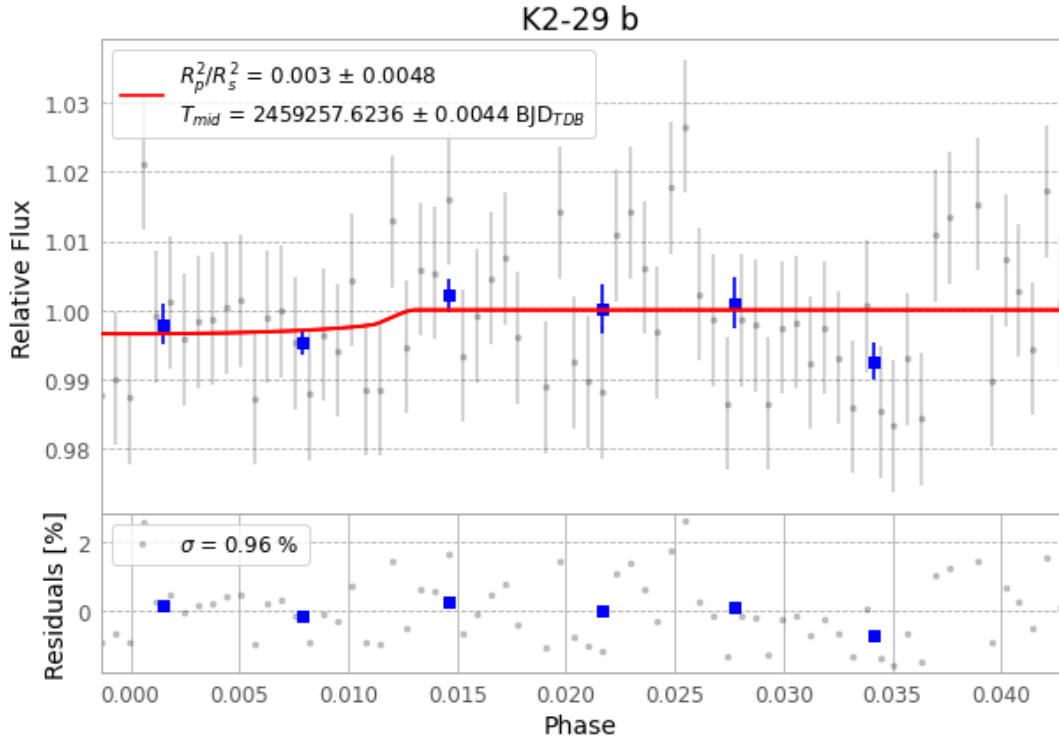


Figure 1 — FinalLightCurve_K2-29 b_2021-02-11.png (SHA-256 a20b49a5c89b58a8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [356, 270], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 0a43d7a9c8fa0524...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

70 FITS frames (46 MB), K2-29210212024812.FITS → K2-29210212061512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-K2-29-210212.log (c0546f85361b...), FinalLightCurve_K2-29 b_2021-02-11.png (a20b49a5c89b...), FinalLightCurve_K2-29 b_2021-02-11.csv (3e39f327bb53...)

Compute resources

Wall time 118.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-16 00:16Z.